COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 1 : Session – 6 : SLO – 2 | Complement arithmetic
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MCQs

• 1. c — 0110 (1010 → 0101 + 1)
• 2. c — 2's complement
• 3. b — 0011
• 4. c — Take 1's complement and mark result negative
• 5. d — Sign of the number

Two-Mark Questions

• 1. 1's complement of a binary number is obtained by inverting every bit (0→1, 1→0). For a 4-bit number, e.g. 1011 → 0100. It represents the negative of the number in 1's complement arithmetic.
• 2. 2's complement subtraction rules: (i) find 2's complement of the subtrahend; (ii) add it to the minuend; (iii) if a carry is generated, discard it — result is positive and final; (iv) if no carry, result is negative — take 2's complement again for the magnitude and attach '-'.

Scenario-Based Long Answer

• A - B = 1011 - 0101 using 1's complement: 1's complement of B = 1010.
• Add: 1011 + 1010 = 1 0101. End-around carry present → add it back to the LSB: 0101 + 1 = 0110.
• Final result: 0110■ = 6■■ (positive, correct: 11 - 5 = 6). Had there been no carry, the result would be negative and interpreted by taking its 1's complement.

Match the Following

• 1. 1's Complement — a) Invert all bits (0→1, 1→0)
• 2. 2's Complement — b) Add 1 to 1's complement
• 3. Carry in 2's complement — c) Ignore and result is positive
• 4. No carry in 1's complement — d) Take 1's complement and add '-'
• 5. Sign-magnitude representation — e) MSB for sign, remaining bits for value

Fill in the Blanks

• 1. 1000 (0111+1) 2. negative (2's complement form) 3. 1111 4. Most Significant Bit 5. 1's complement

Scenario-Based Small Activity

• A - B = 0101 - 0011. 2's complement of B = 1100 + 1 = 1101.
• Add: 0101 + 1101 = 1 0010 → discard carry → 0010.
• Final answer: 0010■ = 2■■ (5 - 3 = 2).

True or False

• 1. FALSE — Flipping bits gives 1's complement; add 1 for 2's. 2. FALSE — 2's complement is used today.
• 3. TRUE 4. TRUE 5. TRUE